

# Harbor's payment QA architecture

A concrete design for AI assistance across a release workflow with 75 offshore QA staff.

75

## offshore QA staff

One payment release.

A platform the next team can reuse.

# The assumed application landscape

Payment APIs offer the first controlled test surface. Other applications follow their own readiness.

## Payment services

Java / Spring Boot  
PostgreSQL journal  
Versioned payment API

### First pilot boundary



## Web + mobile

React journey; existing Selenium and Appium suites

## Provider integration

WireMock during service tests; provider sandbox for integration

## Settlement batch

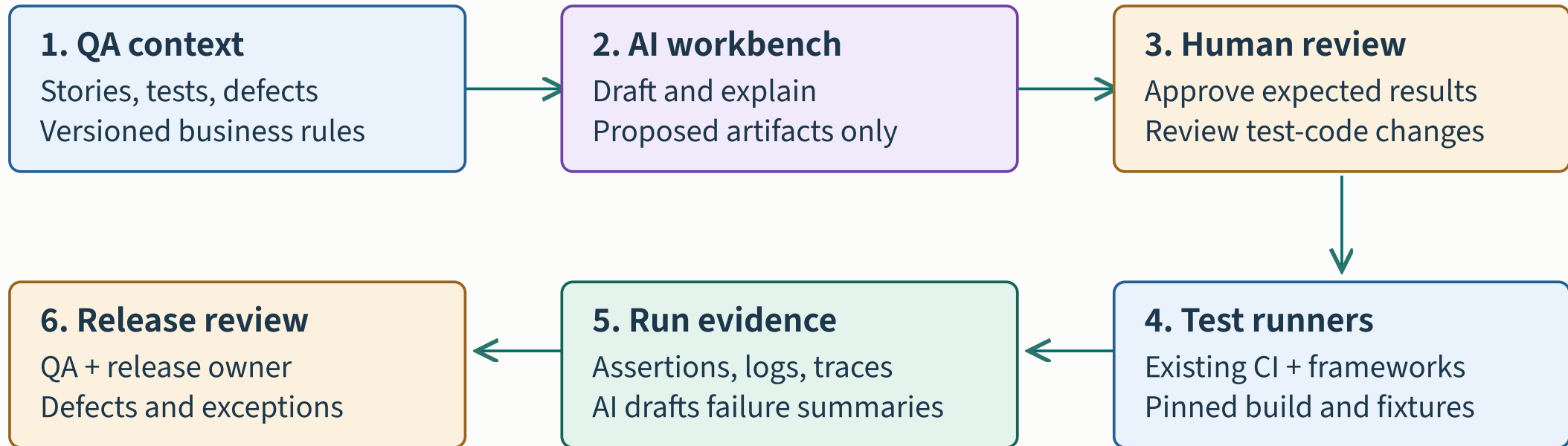
Legacy Java + SQL; shared nightly environment and file reconciliation



Assumed stack: Java/Spring Boot, React, PostgreSQL and Azure. Existing automation remains useful.

# The shared QA platform

The model proposes artifacts. Reviewers approve them. Existing runners produce evidence.



Shared foundation: synthetic data · dependency stubs · artifact IDs · ownership · model configuration

Adapters, context assembly and evidence linking are integration work to build. No turnkey platform is implied.

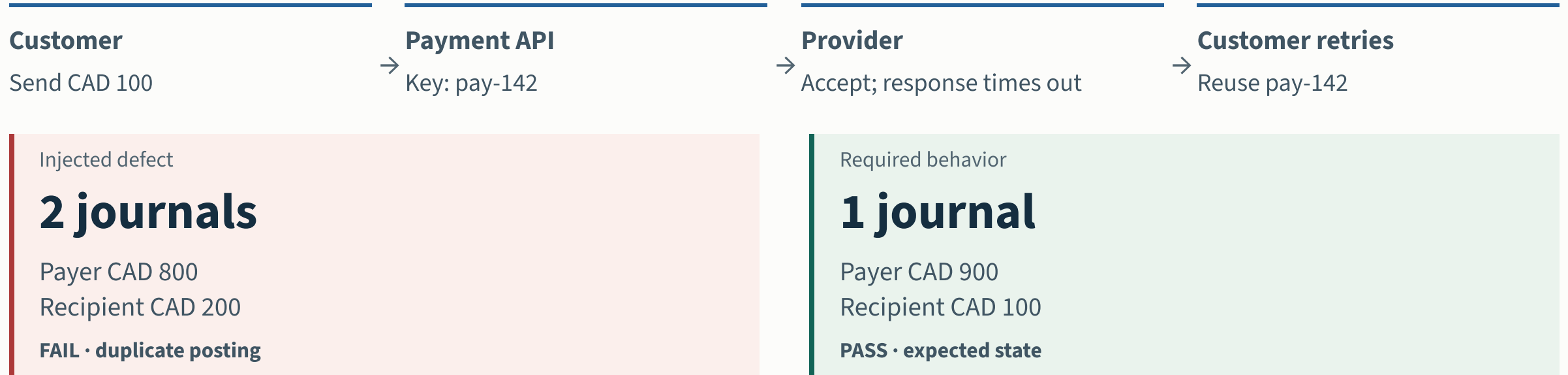
# PAY-142: the payment contract

| CONDITION                                | EXPECTED BEHAVIOR                                                               |
|------------------------------------------|---------------------------------------------------------------------------------|
| Same key + same payload                  | Return the same transfer ID and result. Post the journal once.                  |
| Same key + changed amount                | Reject the conflicting request without another posting.                         |
| Concurrent retries / duplicate callbacks | Converge on one transfer and one balanced journal.                              |
| Expected final state                     | Payer CAD 900.00; recipient CAD 100.00; one journal with two balancing entries. |

Authored business rules: initial balances CAD 1,000.00 and CAD 0.00; transfer CAD 100.00; no fees, FX or other activity. No regulatory claim.

# The retry failure and its detection

The provider accepts a transfer, but its response times out. A retry reaches the posting path again.



Illustrative outcomes for initial balances of CAD 1,000 and CAD 0. A journal contains one debit and one credit. No fees, FX or other activity.

Injecting a duplicate-posting defect should fail the journal-count and balance assertions. A correct success toast cannot override that failure.

# Requirements and test design

PAY-142 starts with an ambiguity: does a timeout mean failure, success or unknown?

---

## Requirements

### AI assistance

Draft ambiguity questions and identify affected payment journeys.

### Output and owner

Versioned acceptance criteria and an open-question log. Product owner + domain QA.

### Human check

Product owner confirms that a retry uses the same key and returns the original transfer.

---

## Test design

### AI assistance

Draft positive, boundary, retry and concurrency cases with requirement links.

### Output and owner

Reviewed test matrix with independently defined expected results. Domain QA lead.

### Human check

Include same-key/different-payload, parallel retry and duplicate callback cases.

Domain QA and the product owner approve expected behavior before generated tests can validate it.

# Test data and environment preparation

Synthetic accounts, known balances and a controlled provider response make the retry repeatable.

---

## Test data

### AI assistance

Draft seed scripts and scenario combinations using synthetic records.

### Output and owner

Versioned fixtures with isolated accounts and known starting balances. Test-data engineer.

### Human check

Verify relationships and exact balances before running tests.

---

## Environment

### AI assistance

Summarize readiness failures and draft dependency stub mappings.

### Output and owner

Reproducible test run with pinned build, data and stub versions. Platform engineer + QA.

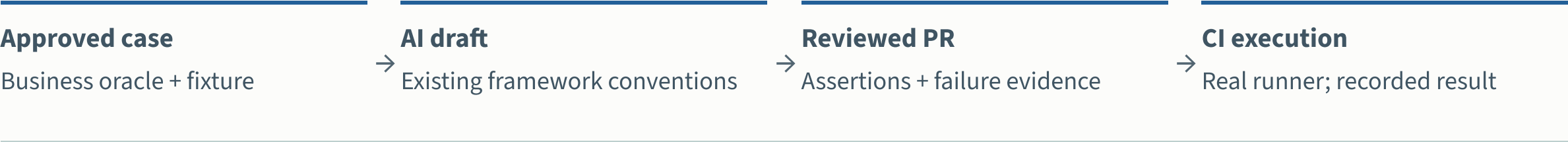
### Human check

A deterministic preflight checks health, schema and fixture readiness.

Container-based service tests do not replace real integration, mobile or batch testing. Validate stubs against the provider contract.

# Generated code enters an ordinary review

Use the approved case matrix and repository examples to produce a candidate test change.



Required assertions for PAY-142

```
same transfer ID on retry
journal count == 1
payer balance == 90000 minor units
recipient balance == 10000 minor units
```

The test must fail when a deliberate duplicate-posting fault is enabled.

The reviewer checks the oracle. A deliberate duplicate-posting fault must fail before the candidate joins the required suite.



# The appropriate test layer

| LAYER                       | EXAMPLE IMPLEMENTATION                                                                                 |
|-----------------------------|--------------------------------------------------------------------------------------------------------|
| Service and API             | REST Assured + JUnit. Cover retry, concurrent requests, idempotency conflicts and journal outcomes.    |
| Web journey                 | Playwright for the new React flow. Check pending state, retry action and the final customer message.   |
| Existing browser and mobile | Keep owned Selenium and Appium coverage. Change frameworks only when maintenance evidence supports it. |
| Batch and integration       | Retain settlement-file reconciliation and provider sandbox checks with explicit scheduling.            |

Test distribution follows behavior and observability. A new browser framework does not solve a shared settlement environment.

# Execution retains required coverage

CI records the build, fixture, stub and test revisions before runners execute the pack.

---

## Execution

### AI assistance

Recommend affected tests and summarize run evidence.

### Output and owner

Runner results, traces and mandatory-suite completion record. Application QA team.

### Human check

Keep the mandatory payment suite while validating selection quality.

AI test-selection recommendations run in shadow mode first. Required payment checks continue on every eligible release.

# Triage preserves the original test intent

A failed journal assertion belongs to the product defect path until evidence establishes otherwise.

---

## Triage & retest

### AI assistance

Cluster related failures and draft evidence-linked defect or repair proposals.

### Output and owner

Reviewed defect disposition, fix and fresh rerun evidence. QA lead + developer.

### Human check

Separate product defects, flaky tests and environment faults. Recheck original intent after a patch.

Capture failed tests, skipped tests and retries. A repair that removes an assertion does not count as a passing outcome.

# A traceable release decision

Every generated artifact and run links back to the approved business behavior.

---

## PAY-142

Approved behavior

---

## Test revision

Reviewed assertions

---

## Run manifest

Build, data, stub

---

## Run evidence

Results + traces

---

## Disposition

Defect / fix / rerun

---

## Release owner

Recorded decision

AI can draft the summary. Missing assertions or unresolved defects remain visible to the release reviewer.

An absent result remains unknown. The release owner resolves exceptions and the QA lead reviews the evidence pack.

# What the demonstration proves

| TRY IN THE CASE EXPLORER | EXPECTED OUTCOME                                                         |
|--------------------------|--------------------------------------------------------------------------|
| Normal transfer          | One posting and expected balances: the modeled checks pass.              |
| Retry with injected bug  | Two postings and incorrect balances: the modeled checks block release.   |
| Retry with deduplication | One posting after two attempts: the modeled checks pass.                 |
| Duplicate callback       | Repeat the event with the same identity and inspect the journal outcome. |

The browser demonstration is a deterministic teaching model. It does not execute REST Assured, call a payment service or prove production correctness.

# Readiness across six dimensions

These dimensions describe the assumed delivery profile. Review the wider dependency register before approving a workflow.

| DIMENSION               | ASSUMED MIXED PROFILE                             | FIRST IMPLEMENTATION PRIORITY                      |
|-------------------------|---------------------------------------------------|----------------------------------------------------|
| Requirements            | Versioned stories; some ambiguity                 | Confirm payment outcomes before generation         |
| DevOps                  | CI available; deployment partly manual            | Pin build and deployment revisions                 |
| Cloud & environments    | Cloud test environment; data reset shared         | Isolate accounts and script data reset             |
| Application testability | Payment API testable; batch dependency shared     | Pilot the API and control provider behavior        |
| Automation              | Java automation usable; UI pack noisy             | Stabilize owned tests and retain required coverage |
| Offshore delivery       | Domain expertise strong; automation skills uneven | Pair reviews and define the next owner             |

Assess each application separately. Unknown infrastructure, data, virtualization, AI access or ownership must be resolved before the affected workflow can proceed.

# Offshore handoff and platform ownership

Each handoff carries a reproducible artifact and a named next owner.

---

Shared overlap window

## Clarify and review

Product owner + QA lead confirm expected payment behavior.

**Handoff: approved case + owner**

---

Offshore execution

## Prepare and run

QA pairs with automation engineers, then runs the pinned build and fixtures.

**Handoff: manifest + evidence**

---

Next owner's workday

## Resolve and retest

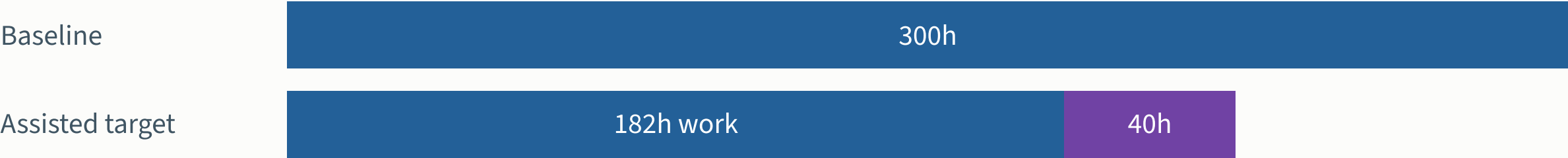
Developers receive a reproducible failure and the assertion that failed.

**Handoff: fix + fresh rerun**

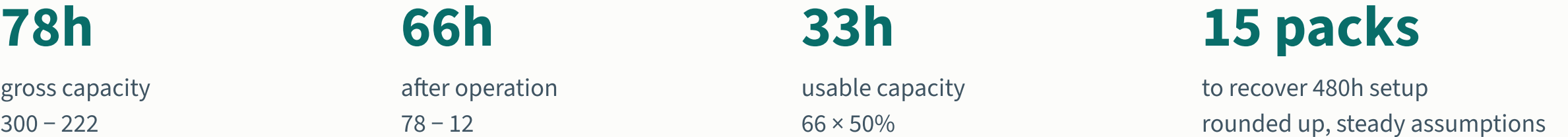
Pilot team: one QA lead, four domain testers, two automation engineers and one data/environment engineer. Product, development and platform support contribute separately.

# Effort includes review and operation

Record time at the eight workflow stages using the same boundary for baseline and assisted packs.



Purple segment = review. Total assisted effort = 222h.



Targets assume verified adoption prerequisites. Negative capacity remains possible; do not extrapolate one pack to all 75 staff or combine overlapping benefits.



# Pilot validation and stop conditions

| STEP           | PROPOSED VALIDATION                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| Design         | Baseline one current pack. Compare at least two assisted packs with similar scope and matched conventional tasks. |
| Record         | Capture task complexity, reviewer time, corrections, failures, waits and environment incidents.                   |
| Stop or repair | Pause expansion for an escaped money-movement defect, unreliable fixtures or tests that pass the injected fault.  |
| Expand         | Require trustworthy mandatory coverage, positive net capacity and a successful second-squad reuse trial.          |

Small pilot samples support a delivery decision with uncertainty. They do not establish a general causal productivity rate.

# Architecture questions for the leads

1. “Can we deploy a known build, reset the data and replay a provider timeout today?”

---
2. “Where do payment outcomes become observable: API, journal, callback or batch file?”

---
3. “Which framework already has maintainers and reliable CI execution?”

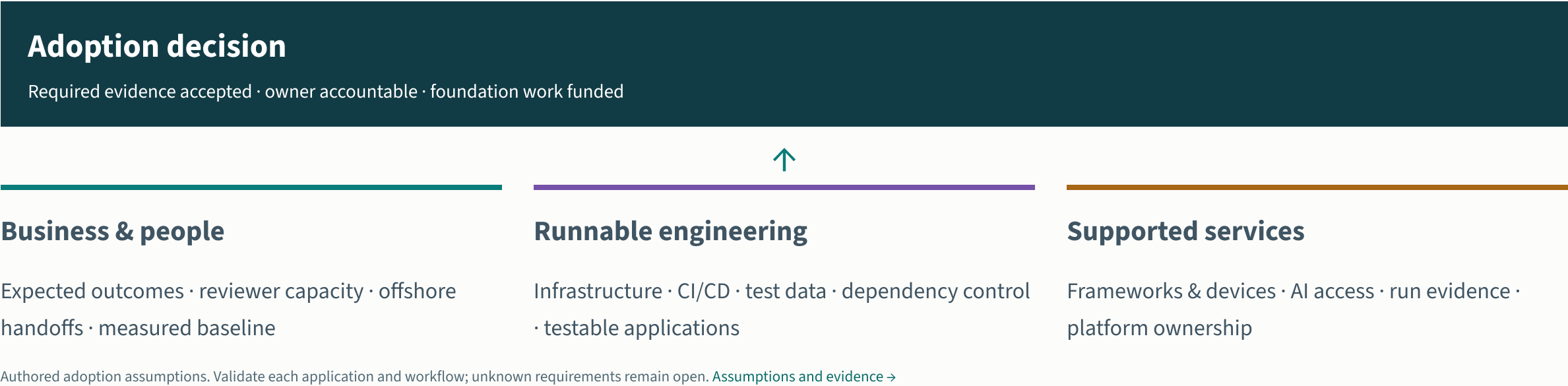
---
4. “Who owns reusable fixtures and who approves the expected financial behavior?”

---

Review the workflow map and a real failed release together. The first implementation backlog should follow those answers.

# Platform prerequisites for the pilot

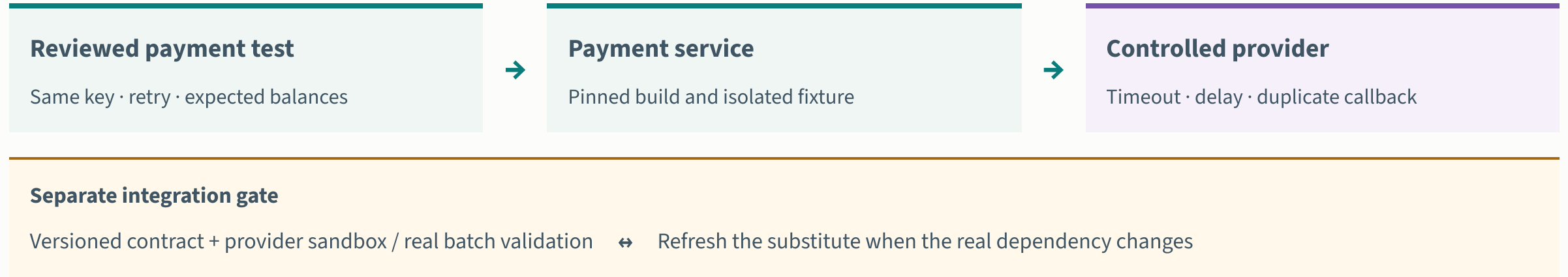
Twelve dependency areas connect the platform to each QA workflow.



Unknown evidence remains a gap. Platform owners, reviewers and supplier leads must accept named actions and review dates before adoption.

# Service virtualization and real integration

Control the payment provider for repeatable tests; keep a separate real-integration gate.



Proposed payment test architecture. A passing substitute is evidence about the modeled behavior, not proof of compatibility with every real dependency. [Dependency strategy and sources](#) →

WireMock is an HTTP example. Contract fidelity, callback behavior and batch access need explicit owners; passing a stub does not prove end-to-end correctness.